Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 · Unit 3 · Formation of PDE

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Formation of PDE

1 Formation of PDE

Study focus: Formation of PDE. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will discuss formation of partial differential equation by elimination of arbitrary functions, solutions of partial differential equations, and solution of first order equations.

Key details

- Remember, there are two ways to form a PDE.
- That is by elimination of arbitrary constants and by elimination of arbitrary functions.
- In a similar manner, a second order equation may arise from an expression containing.
- Two arbitrary functions.
- Remember, it is not always guaranteed that an nth order PDE can be formed from an equation involving n arbitrary functions.
- In some cases, higher order.
- Partial derivatives may be required to eliminate all n arbitrary functions.
- That leads to a PDE of harder higher than n.
- In these cases, the obtain PDE may not be unique.
- For a partial differential equation from an arbitrary function z equals f of x plus y.
- Observe that in this question we have f to be an arbitrary function.
- Because there is only one arbitrary function.
- We proceed by finding first order partial derivatives.
- That's going to be f dash of u times dou by dou x.
- Since u equals x plus y, dou u by dou x becomes 1.
- Because U is a function of X and T.
- So w by dou x becomes f dash of x plus ct plus g dash of x minus c t and w by dou t equals f dash of x plus c t times c plus g dash of x minus ct times minus c because the coefficient of minus c t with respect to t is minus c.

- Since there are two arbitrary functions, we need second derivative also.
- We get f dash of x plus ct plus g double dash of x minus ct.

Example or application from the lecture

- If a given equation contains a single arbitrary function, we can always form a first order PDE.
- Let's take up some of the examples.
- Then the given function becomes z equals f of u.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

In dou square u by dou t square, if you take c square common, we are left out with.

Key details

- F double dash X plus C T plus G double dash X minus C T which is same as dou square u by dou x square hence the second order PDE will be dou square u by dou t square equals.
- C square times dou square u by dou x square.
- Observe carefully in this question we have a single arbitrary function, so we can expect first order PDE and maybe more.
- Differentiate partially with respect to x.
- Observe dou z by doux and double dou z by dou y.
- So eliminate f dash by dividing the first equation by the second.
- From the PDU for this particular function.
- Since z is a function of x and y, we can find out douz by doux and double dou z by dou y.
- That's same as f dash of x square plus y square times 2x plus 1 and dou z by dou y equals f dash of x square plus y square times 2y.
- We cannot solve like previous problem.
- Substitute f dash of x square plus y square in the first expression.
- That leads to dou z by dou x equals 2x of 1 by 2y dou z by dou y plus 1.
- Hence the required PDE will be dou z by dou x minus x by y times dou z by dou y equals one so remember there are two ways to form a pd first one by elimination of arbitrary constants.
- And second by elimination of arbitrary functions.
- Remember, a solution of a PDE is nothing but.
- It is known that a PD can be formed in two ways, one by eliminating arbitrary constants, and second one by eliminating arbitrary functions.
- And these relations are solutions of partial differential equation which is formed.
- There are two types of solutions for partial differential equation.
- Solution containing arbitrary functions and solution containing arbitrary concepts.

Example or application from the lecture

- Let's look at the third problem.

- Given z equals f of x square plus y, where f is an arbitrary function.
- So finally the first order PDE after eliminating f from the given function will be.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

Let's look at what is a complete solution and what is general solution.

Key details

- A solution of a PDE is said to be complete solution if the number of arbitrary consents is same as the number of independent variables present in a PDE.
- Let Z equals f of x plus y.
- We need to form a PDE by eliminating this arbitrary function f.
- It is very easy to see that we can eliminate f dash of x space y by equating dou z by doux and dou z by dou y.
- The equation dou z by dou x minus dou z by dou y equals zero is a partial differential equation formed by eliminating arbitrary function from the original function.
- So an equation containing XYZPQ defines a first order PDE which we have seen already in the session number one.
- A relation of the form phi of UV equals zero where phi is an arbitrary function of u and v.
- and satisfies equation one is called a general solution.
- In a similar manner, the solution obtained by determining the arbitrary function in the general solution by using some specified condition is called a particular solution.
- We will look at the idea of general solution once again in Lagrange's linear equation.
- We understood the idea of solution of PDEs.
- We also learned solution of first order equations.

Example or application from the lecture

- In order to understand solution of a PDE, let's look at an example.
- As we know from previous idea, let us find out douz by doux and dou z by dou y.
- So how do we talk about solution of a PDE through this example?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Formation of PDE transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Formation_of_PDE__d6967667ca1ebee8.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Formation of PDE transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Formation_of_PDE__d6967667ca1ebee8.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Formation of PDE transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Formation_of_PDE__d6967667ca1ebee8.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
Formation of PDE transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/001_Formation_of_PDE__d6967667ca1ebee8.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.

5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00002	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00003	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0052. Sidecar variants compare...
SRC00004	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0052. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00005	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0052. Sidecar variants compare...

C Revision and retrieval practice